

6.3.7 Measurement Process

6.3.7.1 Purpose

The purpose of the Measurement Process is to collect, analyze, and use measurement data to support management and technical decision-making and to objectively demonstrate product and process quality throughout the system life cycle.

6.3.7.2 Outcomes

As a result of successful implementation of the Measurement Process:

- a) Information needs of relevant technical and management processes are identified and prioritized.
 - b) Appropriate measures are identified, defined, or developed to satisfy the information needs.
 - c) Measurement activities are planned, resourced, and conducted in accordance with organizational policies and procedures.
 - d) Data is collected, stored, analyzed, and interpreted using defined procedures.
 - e) Information products are generated and used to support decision-making and provide objective communication to stakeholders.
 - f) The effectiveness of the Measurement Process and the utility of measures are evaluated.
 - g) Potential improvements to the Measurement Process and measures are identified and communicated to the process owner.
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6.3.7.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with applicable organizational policies and procedures.

NOTE 1 — ISO/IEC 15939 provides a more detailed set of activities and tasks aligned with the ones below.

NOTE 2 — Clause 8 of ISO 9001:2000 specifies Quality Management System requirements for measurement and monitoring of processes and products.

6.3.7.3.1 Measurement Planning

This activity defines how measurement will support organizational and project objectives.

- 6.3.7.3.1.1 Describe relevant organizational characteristics that influence measurement (e.g., structure, roles, information flows, project types).
 - 6.3.7.3.1.2 Identify and prioritize information needs for technical and management decision-making.
 - 6.3.7.3.1.3 Select and document measures that satisfy these information needs, including definitions, data sources, units, scales, and responsible parties.
 - 6.3.7.3.1.4 Define data collection, storage, analysis, and reporting procedures.
 - 6.3.7.3.1.5 Define criteria for evaluating the quality and utility of information products and measurement results.
 - 6.3.7.3.1.6 Review and approve measurement plans and allocate required resources.
 - 6.3.7.3.1.7 Acquire and deploy supporting technologies and tools for data management and analysis.
 - 6.3.7.3.1.8 Integrate measurement procedures into relevant organizational and project processes.
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6.3.7.3.2 Data Collection

This activity gathers measurement data in accordance with the defined procedures.

- 6.3.7.3.2.1 Collect data from defined sources using approved methods and tools.
 - 6.3.7.3.2.2 Store data in accordance with organizational data management and security policies.
 - 6.3.7.3.2.3 Verify and validate collected data for completeness, accuracy, and consistency.
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6.3.7.3.3 Data Analysis and Reporting

This activity transforms data into information products that support decisions.

- 6.3.7.3.3.1 Analyze collected data using defined procedures to derive meaningful information products.
- 6.3.7.3.3.2 Interpret analysis results against defined success criteria and decision thresholds.

- 6.3.7.3.3 Document and communicate results to measurement users, decision authorities, and relevant stakeholders in a clear and timely manner.
 - 6.3.7.3.4 Use information products to support technical and management decisions (see 6.3.3 Decision Management Process).
 - 6.3.7.3.5 Ensure traceability of measurement data and results for audit and review purposes.
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6.3.7.3.4 Evaluation and Improvement

This activity assesses the effectiveness of the Measurement Process and identifies improvements.

- 6.3.7.3.4.1 Evaluate the quality, timeliness, and usefulness of information products and measurement results.
 - 6.3.7.3.4.2 Evaluate the performance and adequacy of the Measurement Process itself against defined criteria.
 - 6.3.7.3.4.3 Identify and communicate potential improvements to the process owner or designated authority.
 - 6.3.7.3.4.4 Implement approved improvements in accordance with organizational policies.
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Cross-References

This process supports and is supported by:

- **Quality Management Process (6.2.5)** — for defining quality objectives and using measurement for quality monitoring.
- **Decision Management Process (6.3.3)** — for providing objective data to support trade-offs and selections.
- **Configuration Management (6.3.4)** — for measuring baseline integrity and change impacts.
- **Verification (7.2.4)** and **Validation (7.2.5)** — for collecting objective evidence of compliance.
- **Review (7.2.6)** and **Audit (7.2.7)** — for assessing measurement implementation and results.